nums = [2, 7, 11, 15], target = 9

0 1 2 3

i j i j

Brute Force

```
for (int i=0; i<n-1; i++) {          ⟶ O(n-1)
    for (int j=i+1; j<n; j++) {      ⟶ O(n-1)
        if (num[i] + num[j] == target)
            return {i, j}
    }
}
return { };
```

(n-1) * (n-1) ~ (n-1)²

∴ O(n-1)² ↓

O(n²)

O(n²)

O(n)

(*) Two - Pointer Approach :

target (9)

2 , 7 , 11 , 15

2+15=17

2+11=13

left

right

O(n)

while(l<r)

sum

if (nums[left] + num[right] == target)

return {left, right};

else if (sum > target)

right --;

else

left ++;

}

left = 0

right = 1

left == right

Array

1. Two-Pointer
2. Sliding window
3. Prefix sum
4. Kadane's Algorithm

Hash_map

arr = (key : value)

2, 7, 11, 15



7 : 1

11 : 2

5 : 3

vector<<pair<int,int>>

arr

arr.push_back( )

Sugar → 25% Price like.

100 unit ← Consumption ↓

Expenditure (constant)

1. Unitary method
2. Ratio-method
3. Formula method

Before

P	C
100	100

After

P	C
125	(100 − 80)
20%	20 unit

125 টাকায় → 100 unit

1 " → $\frac{100}{125}$ "

100 টাকায় → $\frac{100 \times 100}{125}$

$= \frac{400}{5} = 80$ unit

Price, Consumption & Expenditure

20% → $16\frac{2}{3}$% → 16.66%

$100 = 26 \times 5$

$E = P \times C$

$P \times C = E$ (const.)

$P \uparrow \; C \downarrow$

$P \downarrow \; C \uparrow$

25% → N, D

P_1 4 → 5 (C1)

P_2 5 → 4 (C2)

$P \propto \frac{1}{C}$

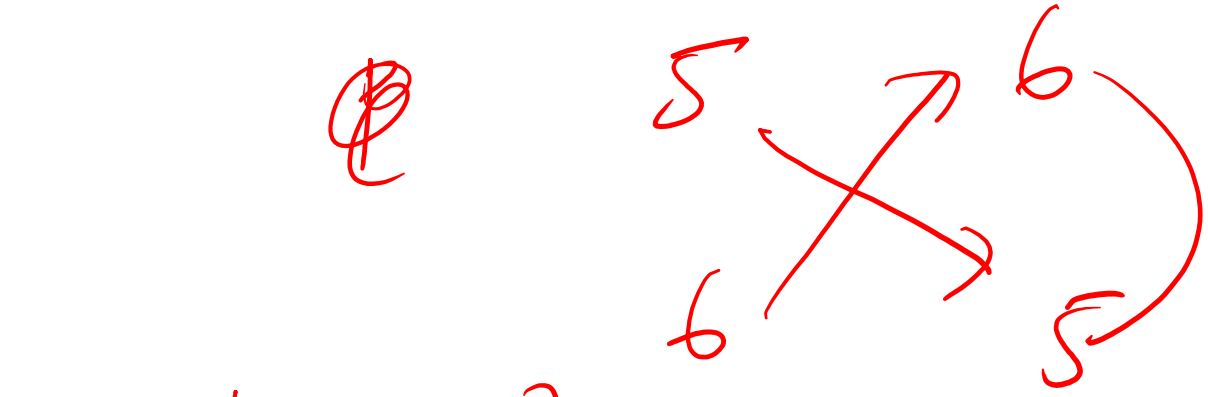
$\frac{P_1}{P_2} = \frac{C_2}{C_1}$

$\frac{1 \text{ unit}}{5 \text{ unit}} \times \frac{20}{100} = 20\%$

Price

20% Hike

20% = $\frac{1}{5}$



$\frac{1}{6} \times 100$

$= \frac{100}{6} = 16\frac{2}{3}$

$16\frac{4}{6} = 16\frac{2}{3}\%$.

6) 100 (16
-96

4

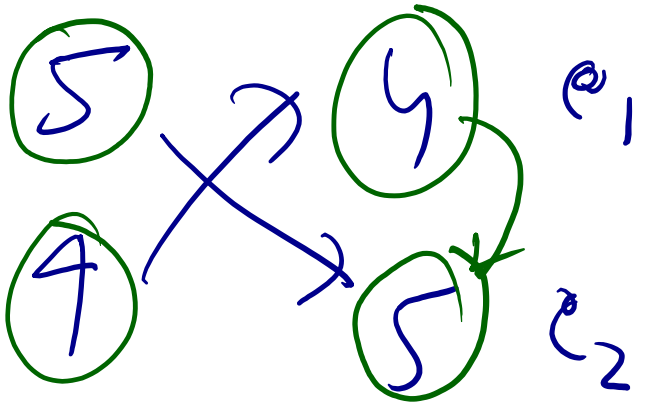
Price

20% ↓

Consumption ↑

Exp. Cost.

20% = 1 → N
5 → D



$\frac{1}{4} \times \frac{50}{100} = 25\%$

3. formula

Price hike

$$\left[\frac{r}{100+r} \times 100\right] \% \Big/ \left[\frac{r}{100-r} \times 100\right]$$

$$\frac{20}{80} \times \frac{5}{5} \times 100$$

$$= 25\%$$

25%↑

$$\frac{25}{100+25} \times 100$$

$$= \frac{\cancel{25}\,5}{\cancel{125}\,5} \times \cancel{100}\,4 = 20\%$$

P = 20% ↓ Sugar

C = 2 kg ↓ 160 ₹

i) original consumption

ii) New "

iii) original / Actual cost Per kg

iv) New / Actual cost Per un

P = 20% ⇒ a

C = 5% ⇒ b

P x C = E

↑ ↑ = ↑

net effective %

Successive

(*)

$$\left(a+b+\frac{ab}{100}\right)\%$$

20 + 5 * 1

= 26

$$81x = \frac{81}{100}$$

$$\frac{81}{181} \times 100$$

$$181 \overline{)8100} (4$$

$$-724$$

$$860$$

$$3 \quad 181 \quad \times 4 \quad -724$$

$$100 \quad 181 \quad 181 \quad 100$$

$$2$$

$$181 \quad \times 3 \quad 543$$

20% ↓

50kg

450 ₹

20% = $\frac{1}{5}$

P → 5 : 4

C → 4 : 5

9 450 ₹ — 9 ₹

250 ₹ — 5) 9 (1

1 unit → 50 kg

4 unit → 200 kg.

9 450 ₹ — 9 ₹

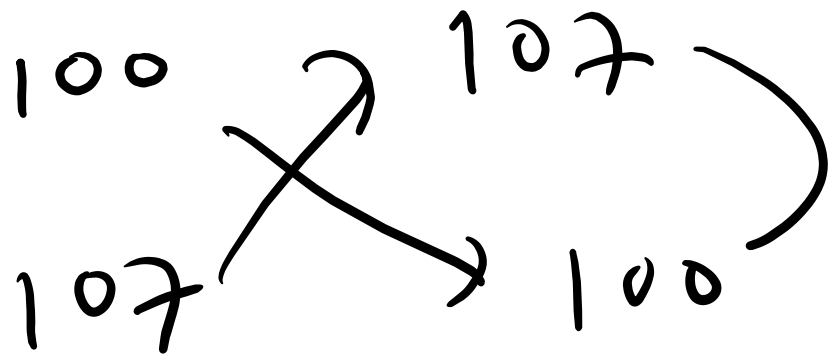
270 ₹ — 4) 9 (2

8

1

8

$7\% = \frac{7}{100}$



$\frac{7}{107} \times 100$

$107 \overline{)700} (6$

4
107
× 6
642